



Investigation report

Haul truck and light vehicle collision, Bindarra Gold Mine - AAIB-style causal and contributory factors report

Drafted 04/10/2026

INVESTIGATION WORKFLOW SUITE

Contents

Synopsis.....	3
Factual information	4
History of the occurrence.....	4
Recorded information.....	5
Equipment information	5
Meteorology.....	5
Personnel information	5
Site information.....	6
Organisational and management information	6
Tests and research.....	7
Other information	7
Analysis	8
External Influences	8
Organisational Influences.....	9
Risk Controls.....	11
Local Conditions	13
Individual Actions	14
Occurrence Events.....	16
Conclusions	19
Findings	19
Causal factors	21
Contributory factors	22
Other findings.....	22
Safety action and recommendations.....	23
Traffic management guidance leaves the review trigger undefined and the interaction zones to the site	23
Access spur cut and opened to traffic under a works order, with no management of change raised	23
Contractor induction accepted as equivalent, with the site supplement and pit driving assessment never verified.....	24
Positive radio calls required only at listed intersections, and the list is rewritten once a year	24
Collision awareness alerts suppressed inside a geofence extended to cover the new intersection ..	25
One emergency response paramedic rostered on night shift, with the second responder called from the village	25
Distribution	26

Synopsis

At 02:47 on 12/08/2026, in the Stage 4 open pit at Bindarra Gold Mine, an empty 220 t class rigid haul truck descending the main ramp struck the right side of a light vehicle emerging from the RL 312 dewatering spur at 24 km/h. The light vehicle was pushed 18.4 m down the ramp and rolled onto its left side against the inner windrow. Its driver, a grade control technician employed by a survey subcontractor on his third night shift at the mine, sustained a fractured left clavicle, three fractured left ribs and a small pneumothorax. He remained conscious, was extricated through the rear of the vehicle 27 minutes after the collision, and was flown to the metropolitan trauma centre. The light vehicle was destroyed. Its seatbelt was found fastened and loaded, and its roll-over protection structure deformed without being breached. The haul truck was damaged at the front left and returned to service on 14/08/2026. The Stage 4 pit stood down for 9 hours 15 minutes, and the main ramp below RL 340 was closed for 22 hours.

It was very likely that neither operator was aware of the other on reaching the intersection, which led directly to the collision. The spur joined the ramp beyond a crest, giving 41 m of sight line against a plan minimum of 120 m and a stopping distance of 96 m, which very likely contributed. The spur had been cut under a works order without a management of change record and so never reached the traffic management plan, the in-cab map or the collision awareness configuration, which very likely contributed. The plan required positive radio calls only at listed intersections, a list changeable only at the annual review, which very likely contributed. That neither operator made a call likely contributed. The collision awareness system detected the light vehicle but suppressed the alert inside a geofence extended in June 2026. The alert would have come after braking began. Its likely influence lay in months of silence teaching the truck operator that silence meant an empty road. The light vehicle entered the ramp without stopping or establishing the truck's position, which very likely contributed, although a full stop would have gained only a few metres of sight line. The driver's site supplement and pit driving assessment had never been verified, which likely contributed. The width of discretion in Mines Safety Guideline MSG-14 likely contributed, a finding reached on balance and held with less confidence than the others. The mine was not in breach of the guideline.

The rostered night shift emergency response arrangement and the truck operator's position on night 10 of 14 were not found to have influenced the occurrence, but both increased risk. A wet ramp surface and degraded truck brakes were ruled out. Whether dust reduced the truck operator's view could not be determined, as the accounts differ and nothing at the site measures visibility. The truck operator braked and steered towards the wall within 1.4 seconds of sighting, reducing the impact speed from 37 km/h to 24 km/h and keeping the light vehicle against the inner windrow.

At 05:10 on 12/08/2026 the Mining Manager directed that a positive call be made before entering any Stage 4 intersection. The spur was closed. Survey measured the ten other unrefereed Stage 4 roads, and two were closed pending a stop control. Warrigal technical services began reporting suppressed proximity events. Recommendations address referral of new roads under the traffic management plan, the management of change exclusion for earthworks, and control of suppression geofence boundaries.

Factual information

History of the occurrence

The RL 312 dewatering spur is a 240 m access road in the Stage 4 open pit at Bindarra Gold Mine, cut under works order WO-26-4471 to reach dewatering bore DW-14. The earthworks were recorded complete on 31/07/2026. On 02/08/2026 the bore was energised and light vehicles first used the spur. The Stage 4 works log records that the construction road closed sign at the spur mouth was removed that day and nothing replaced it.

At about 22:10 on 04/08/2026 a dewatering technician drove out of the spur in front of a descending haul truck, which he estimated at about 60 m away, then reversed and let it pass. At 23:48 he submitted hazard report HAZ-2026-0917, which included:

- "The problem is you cannot see up the ramp from that road. There is a rise in the ramp and a windrow along the side and you are out on the road before you can see anything."
[Hazard report HAZ-2026-0917]

The Dewatering Superintendent closed the report on 06/08/2026, recording that the crew had been reminded at the pre-start on 05/08 to take care entering the main ramp and to give way to haul trucks. The risk rating was reduced from Medium to Low, and the report was not referred outside the dewatering department.

On 09/08/2026 the light vehicle driver, a grade control technician employed by Merrivale Survey & Geotechnical, was issued a site access card at the Bindarra gatehouse and worked his first night shift there that evening.

On the night of 11 to 12/08/2026 dispatch called the Stage 4 crews back from crib at 02:22:41. Haul truck HT-407, on its second cycle after crib, reported dumped and coming back down at 02:38:55.

The driver completed levels at the DW-14 collar and drove LV-118 out along the spur. He recalled leaving at about ten to three, later than the recorded collision time. From the spur length and his position at the collision, the investigation estimated that he left at about 02:45 and reached the spur mouth at about 02:47. He stated that he slowed, looked left up the ramp, saw nothing and drove out at about 25 km/h, and that he did not believe he came to a complete stop. Nobody on the shift knew LV-118 was in the pit.

HT-407 descended empty at 41 km/h at RL 368 and 37 km/h at RL 314. Its operator first saw the light vehicle at about 40 m, braked and steered right towards the windrow. At 02:47:12 the truck's front left struck the right side of LV-118 at 24 km/h, 6.2 m into the ramp from the eastern edge of the spur mouth. The light vehicle was pushed 18.4 m down the ramp, went onto its left side 11.2 m from the impact position and came to rest against the inner windrow, facing down-ramp. HT-407 was stationary at 02:47:14.3, 19 m below the intersection. Neither operator made a radio call before entering the intersection.

The truck operator reported the collision on the pit channel at 02:47:41. At 02:48:01 dispatch stopped all Stage 4 units and told him to stay in his cab, releasing him at 02:48:44 once the ramp was clear. The emergency response team was paged at 02:48. The first response vehicle entered the pit at 02:51 and reached the scene at 02:54. The second responder arrived from the accommodation village with the hydraulic rescue tool at 03:07. The driver, conscious throughout, was extricated through the rear of the vehicle at 03:14, reached the surface medical facility at 03:41 and was flown to the metropolitan trauma centre at 07:40. He sustained a fractured left clavicle, three fractured left ribs and a small pneumothorax.

The Stage 4 pit stood down for 9 hours 15 minutes and the main ramp below RL 340 was closed for 22 hours. The occurrence was notified to the State Mines Inspectorate as a reportable injury.

Recorded information

Recorded times are in UTC. Local time is UTC plus 8 hours and is used elsewhere in this report.

The HT-407 fleet management system download (WMS-TS-2026-114) gave a last steady data point at 18:47:06, at 37 km/h at RL 314, 41 m above the intersection. At 18:47:07.4 service brake pressure rose from 0 to 620 kPa in 0.4 seconds, the retarder went to maximum and 14 degrees of right steering was applied. Speed was 32 km/h at 18:47:09.2 and 24 km/h at the collision at 18:47:12.0. Mean deceleration from brake application to collision was 1.85 m/s². LV-118 has no fleet management system, and its collision awareness transponder is a passive tag.

The HT-407 collision awareness system event log recorded the following, 2.8 seconds before the collision and 1.8 seconds after braking began. Nothing was presented in the cab.

- "18:47:09.200 PROXIMITY HT-407 tag LV-118 range 21.4 m closing 13 km/h zone S4-LOAD alert SUPPRESSED (geofence)" [Collision awareness system event log, HT-407]

Pit channel recording RAD-2026-0332, timed by the repeater clock (checked to the fleet management system within 0.4 s), contains no transmission from HT-407 between 18:38:55 and 18:47:41, and none from LV-118. The first report of the collision was:

- "I've collided with a light vehicle at the three-twelve spur. He's rolled. I need ERT and I need them now." [Pit channel recording RAD-2026-0332, 18:47:52]

The driver stated that his radio was on channel 12, as the dewatering foreman had directed, and that he did not remember hearing anything on it. [NOT ESTABLISHED: whether channel 12 was the Stage 4 pit channel]

Equipment information

HT-407 is a 220 t class rigid haul truck, empty at the time. Its collision awareness system operated at Level 7, operator advisory. Level 9, automated intervention, is available on this fleet but not enabled at Bindarra. An advisory alert is raised when a tag enters a 60 m detection ring at a closing rate above 10 km/h, except inside a suppression geofence. Damage was confined to the front left bumper, the left mudguard and the lower two access ladder steps. The truck returned to service on 14/08/2026.

LV-118 was destroyed, with damage concentrated on the right side forward of the B pillar. The roll-over protection structure was deformed inward at the right rear mounting but not breached. The driver's seatbelt was found fastened, with loading marks on the webbing. The 4.5 m amber beacon whip was sheared at its base.

Meteorology

The ramp surface was dry. The truck operator said some dust was hanging and he could not say whether it made a difference. The light vehicle driver was firm that the air was clear, that he could see the ramp and the windrow, and that there was no dust in the spur. [NOT ESTABLISHED: any measured or independently observed dust or visibility condition at RL 312 that night]

Personnel information

The truck operator had 11 years in open pit mining, 7 on 220 t class haul trucks and 14 months at Bindarra. His verification of competency on type, issued in 06/2025, was current, and he completed his annual refresher in 02/2026. He had received no traffic management refresher since his site

general induction in 06/2025. He was on night 10 of a 14 night swing, 10 hours 47 minutes into a 12 hour 30 minute shift, having slept 6 hours 30 minutes in the preceding day. He described himself as tired in the way usual on a tenth night, but not falling asleep.

The night shift supervisor had 19 years in open pit mining and 2 years 6 months in the role, on the same shift pattern as his crews.

The light vehicle driver had 11 years in survey, 6 in mining, and 4 years with Merrivale. He completed Merrivale's two hour online general mining induction on 24/07/2026. His card was issued after a 20 minute gatehouse briefing on muster points, speed limits and personal protective equipment. There is no trace that he received the Bindarra site-specific supplement or completed the Bindarra light vehicle pit driving assessment. The collision occurred on his third night shift at Bindarra.

The truck operator and the supervisor tested negative for drugs and alcohol. [NOT ESTABLISHED: whether the light vehicle driver was tested]

Site information

At RL 312 the main ramp descends at 9.1 per cent and is 24 m wide between windrows. The spur joins from the east, on the inside of a left-hand bend, at 71 degrees and 34 m beyond the high point of a crest. The survey found that a descending truck operator at an eye height of 5.80 m first sees the spur mouth at 41 m, and a driver at the spur mouth at an eye height of 1.45 m first sees a descending truck at 38 m. The 1.6 m inner windrow screens the ramp for the last 25 m of the spur's approach. There was no warning sign on either ramp approach, no give-way or stop control, no delineation and no posted speed for the spur.

Organisational and management information

Clause 6.2 of the Bindarra traffic management plan, BIN-TMP-004, requires intersection sight distance "not less than the stopping distance of the largest vehicle using the road at the posted speed, and in no case less than 120 m". Clause 6.4 requires warning signs, give-way control and delineation, but only at intersections on the plan's road layout drawing, which did not include the spur. Nor was the spur among the nine Appendix B intersections requiring a positive radio call. All three interviewees independently described those nine as the whole of the calling practice, and the supervisor had never corrected anyone for not calling at an unlisted intersection.

The plan was reissued after annual reviews on 14/11/2023, 10/11/2024 and 12/11/2025. In its eleven year history no out-of-cycle review has been carried out under clause 9.1. Since the last review 14 roads have been cut in Stage 4. Three were cut under a management of change record and referred to the plan owner. The other 11 were cut under works orders and none was referred. WO-26-4471 classified the spur as "Routine earthworks within the approved Stage 4 design" with "Management of change not required, CRL-PRO-022 clause 5.3", and its handover field read "N/A". Publishing a revised in-cab road network is a survey task raised by a management of change record. The in-cab map, last published on 24/07/2026, did not show the spur.

The S4-LOAD suppression geofence was drawn on 14/03/2025 around the loading face to stop nuisance alerts. On 09/06/2026 the contractor's technical services group extended it northward, placing the intersection inside it. The extension was not recorded in, or reviewed against, the plan. No one on site held training on the geofences. The supervisor knew of suppression at the dig from mid 2025 but not of the extension. The truck operator had never received an alert in the pit in 14 months and learned of the suppression the day after the collision. The fleet's standing alert report does not include suppressions.

Contractor Management Standard CRL-STD-009 clause 7.4 states that the site supplement and pit driving assessment "shall not be substituted", and clause 7.5 makes the engaging department responsible for verifying them before a card is issued. The gatehouse issues cards against the recognition agreement alone. There is no permit, call-in, board or check for a light vehicle entering an operating pit.

The night shift has one rostered mines rescue paramedic. A second responder, required before an extrication, is called from the village 11 km from the gatehouse. The post-incident review found the extrication was delayed by approximately 9 minutes waiting for the second responder and the tool, and that the delay did not affect the clinical outcome on this occasion.

Tests and research

HT-407 achieved 1.9 m/s² in its monthly brake test on 03/08/2026, and the retarder and service brakes performed as specified throughout the event. Without braking, the collision speed would have been 37 km/h. The calculated stopping distance of an empty 220 t class truck at the posted 40 km/h on a 9 per cent descent, with a 1.5 second response, is 96 m. The LV-118 beacon lamp worked when tested after recovery.

Other information

A search of Bindarra hazard reports in the 24 months to 11/08/2026 found three earlier reports, from 03/2025, 09/2025 and 02/2026, each describing a light vehicle and a haul truck meeting unexpectedly at an unlisted intersection, at three different locations. Like HAZ-2026-0917, all were closed within three days with a reminder to take care. None was referred to the plan owner or led to a plan change or physical control.

At 05:10 on 12/08/2026 the Mining Manager directed that a positive call be made on the pit channel before entering any Stage 4 intersection, listed or not.

Analysis

External Influences

Traffic management guidance leaves the review trigger undefined and the interaction zones to the site

The traffic management plan at Bindarra was written to satisfy Mines Safety Guideline MSG-14 revision 4, issued by the State Mines Inspectorate in 2021 under regulation 6.22 of the Mining Operations Safety Regulations 2018. The regulation itself requires only that "the operator shall manage the risks arising from the interaction of vehicles at the mine". The guideline is one way of discharging that duty, and it sets the framework within which the operator wrote its own controls.

Three of its provisions are relevant here. Clause 3.2 requires the plan to be "reviewed at intervals not exceeding twelve months, and following any significant change to the mine's road network, vehicle fleet or operating pattern". The guideline does not define "significant change" and gives no example of one. Clause 5.1 requires the operator to "nominate the intersections, crossings and other interaction zones at the mine at which positive communication between vehicles is required before entry". It does not say on what basis the nomination is to be made, and it sets no minimum. Clause 5.4 allows detectability of a light vehicle to be provided by "line of sight, elevated marking, radio communication, a proximity detection system, or a combination of these", without saying which means, or how many, must be provided. Under clause 3.5 the plan "is a site document" and is not approved by the Inspectorate or submitted to it except on request after an occurrence or an inspection. The text of the guideline is documentary and not in dispute. It is virtually certain that these provisions existed in this form.

On the day of the occurrence Bindarra's plan met the guideline on its face. It had been reviewed within twelve months, it nominated nine interaction zones, and it provided for detectability by line of sight and by elevated marking. The mine was not in breach of the guideline or the regulation. This analysis does not suggest any failure by the Inspectorate. Its concern is the extent of the discretion the guideline leaves to the operator, and how that discretion was exercised at Bindarra.

At Bindarra all three open terms were filled in by a single document, BIN-TMP-004, which was reviewed once a year and seen by nobody outside the mine. Clause 9.1 of the plan repeats the guideline's review trigger without defining it. Clause 4.3.1 requires a positive call only at intersections listed at Appendix B. According to the note to that appendix, an intersection is added only where it "appears on the road layout drawing at Schedule 1" and the annual review assesses it as needing positive communication. Schedule 1 is reissued only at the annual review. No provision allows an intersection to be added between reviews, and no provision requires a new road to be referred to the plan's owner when it is cut. The document register shows annual reviews on 14/11/2023, 10/11/2024 and 12/11/2025, and no out-of-cycle review in the plan's eleven years. Since the last review, 14 new roads had been cut in the Stage 4 pit. The three cut under a management of change were referred to the plan's owner. None of the 11 cut under a works order was referred, and the RL 312 spur was one of those 11. In the 24 months to 11/08/2026, four hazard reports described a light vehicle and a haul truck meeting unexpectedly at an intersection not listed at Appendix B. Each was closed within three days with a reminder to the crew, and none was referred to the plan's owner.

The argument for influence is that a defined trigger and a stated minimum would have given those eleven roads a route into the plan. Each internal control that failed in this occurrence failed for want of exactly that route.

The argument against is also substantial. The guideline did not prevent anything. Bindarra could have held an out-of-cycle review under its own clause 9.1 at any time, and MSG-14 does not stop a mine from listing every intersection it has. On this view the mine did not fail to act because the guideline was open. It failed to act because nothing inside the mine put the question in front of the plan's owner, and the external provision is background only.

That argument was weighed and not accepted as the whole explanation. The internal arrangements are those found to have existed under the works order process and the Appendix B listing, which are argued in the Organisational Influences and Risk Controls subsections. Those arrangements were built to fill a discretion that the guideline left open, and they filled it with a single annual loop that had no mechanism for referral. Even so, the influence of the guideline is put no higher than likely. That is less firm than the other influence findings in this report, and it rests on a judgement about how the guideline shaped the site's arrangements rather than on direct evidence that it did.

It is concluded that the latitude MSG-14 left to the operator in defining a significant change, nominating interaction zones and providing detectability likely contributed to the occurrence, and it is treated as a causal factor, with the qualification that the evidence on its influence was finely balanced.

Organisational Influences

Access spur cut and opened to traffic under a works order, with no management of change raised

Clause 5.2 of CRL-PRO-022 requires a management of change record "before any of the following: a change to the mine's road network". Clause 5.3 then provides that clause 5.2 "does not apply to routine earthworks carried out under a works order within an approved pit design, where no new plant or vehicle type is introduced."

Works order WO-26-4471 was raised by the Mining Engineer on 28/07/2026 and approved by the Mining Superintendent on 29/07/2026. It cut 240 m of access road to bore DW-14 at RL 312. It classified the work as "Routine earthworks within the approved Stage 4 design" and recorded "Management of change not required, CRL-PRO-022 clause 5.3."

The procedure, the works order and the survey department's road register agree with each other. It is virtually certain that the spur was built and opened to traffic without a management of change record.

What mattered was what such a record would have carried. Under clause 5.4, the impact assessment refers the change to three parties:

- the traffic management plan owner;
- the survey department, for the fleet management system road network;
- the contractor's technical services group, for the collision awareness system configuration.

Those referrals exist only inside the management of change record. The works order form at Schedule 4 has no equivalent fields. WO-26-4471 carried no scope for signage, survey pickup, referral or road network update. Its handover field was the only place any of these could have been recorded, and it read "N/A".

Each of the three paths was found untouched on the night:

- **In-cab map.** The road network had last been published on 24/07/2026, before the spur was cut, and showed the main ramp with no intersection at RL 312.

- **Traffic management plan.** The spur was not on the plan's road layout drawing. Clause 6.4 therefore did not apply to it, so it required no advance warning signs, no give-way control and no delineation. None was installed when the construction sign was removed on 02/08/2026. The intersection was also not among those at which the plan requires a positive call.
- **Collision awareness system.** The suppression geofence already enclosed the location and was not reviewed against the new intersection.

This was not an isolated case. Since the 12/11/2025 plan review, fourteen new roads were cut in Stage 4. The three raised under a management of change record were referred to the plan's owner. None of the eleven raised under a works order was. The exclusion at clause 5.3 was applied as written, and its effect was to remove every consequence of putting a new intersection into an operating haul road.

The consideration against this factor is that a referral is not itself a control. Had a record been raised, the plan's owner might have considered the spur and still left it unlisted. That possibility does not displace the conclusion, because the management of change record was the only route by which the intersection could have reached the plan, the in-cab map or the geofence configuration. It is very likely that its absence contributed to the occurrence. It did so through the sight line shortfall and the positive call list, which are argued under Local Conditions and Risk Controls. The guidance framework above this factor is discussed under External Influences.

The evidence places the absence of a management of change record for the RL 312 spur as a causal factor in the occurrence.

Contractor induction accepted as equivalent, with the site supplement and pit driving assessment never verified

Clause 7.3 of CRL-STD-009 permits a contractor's general induction to be accepted as equivalent where the contractor holds a current recognition agreement. Merrivale Survey & Geotechnical had held one since 2023. Clause 7.4 is explicit that equivalence "does not extend to the Bindarra site-specific supplement, or to the Bindarra light vehicle pit driving assessment, which shall be completed at site by every person who will drive a vehicle in an operating pit and shall not be substituted." Clause 7.5 makes the engaging department responsible for verifying this "before a site access card is issued."

The standard is sound. The failure lay at the point where it was meant to be applied. The gatehouse issued cards against the recognition agreement alone. It held no record of either item and was not asked for one. Nothing in the access card system distinguished a person who may drive in an operating pit from one who may not.

The driver's training file showed two items:

- the Merrivale two hour online module, completed on 24/07/2026;
- a 20 minute gatehouse briefing on 09/08/2026, covering muster points, speed limits and personal protective equipment.

It showed no site supplement and no pit driving assessment. The driver had not seen BIN-TMP-004 and did not know it existed. The occurrence took place during his third night shift at the mine.

It is extremely likely that he was driving in an operating pit without either item. Some doubt remains because Merrivale's own records for the site supplement were not available.

This is not a question of the driver's conduct. He held a card issued through the process the site operated. That process had no place to record the verification its own standard required.

The influence of this factor is more qualified. The pit driving assessment is where a light vehicle driver is taught how the pit's traffic rules work in practice, including give-way, positive calls and what to do at a blind intersection. Two considerations weigh against a stronger conclusion. First, the rule he would have been taught, that light vehicles give way at all times, was one he was already applying. Second, the 41 m sight line at the intersection would defeat attention on its own, whatever the training.

On balance, it is likely that the missing assessment contributed to the light vehicle entering the ramp without stopping and without establishing the truck's position. That action is argued under Individual Actions.

The evidence places the unverified contractor induction as a causal factor, held at a lower confidence than the absence of a management of change record.

Risk Controls

Collision awareness alerts suppressed inside a geofence extended to cover the new intersection

The collision awareness system on HT-407 operated at Level 7, operator advisory. Level 9 intervention was available on the fleet but was not enabled at Bindarra. Advisory alerts were raised when a tag entered the 60 m ring at a closing rate above 10 km/h, except inside a suppression geofence.

The S4-LOAD geofence was drawn around the loading face on 14/03/2025 to stop nuisance alerts. The investigation accepted that purpose as legitimate, because an alert rate high enough to make the system unusable would defeat it altogether. On 09/06/2026 the contractor's technical services group extended the geofence northward as a system configuration task. Its boundary then lay 180 m up-ramp of the RL 312 intersection. The extension was not recorded in the traffic management plan and was not reviewed against it. No training on the geofences is recorded for any person on site.

The event log shows that the system detected LV-118 at 02:47:09.2 at 21.4 m, closing at 13 km/h, and suppressed the alert. It is virtually certain that the suppression existed and covered the intersection, since the configuration, the log and the twelve month statistics came from the system itself and were downloaded sealed.

The evidence on influence points two ways, and both readings are set out here. Against influence, the suppressed alert would have sounded 1.8 seconds after the operator had begun braking and 2.8 seconds before impact. It would have told him nothing he did not already know. The system could only advise, and with 41 m of sight line the outcome was fixed before either the alert or the sighting. On that reading the suppression is a serious matter for the fleet but played no part in this collision.

The case for influence does not rest on the final seconds. In the twelve months before the occurrence, HT-407 raised 412 advisory alerts, all outside the geofence and none inside it. Over the same period, 68 proximity events inside the geofence were suppressed across the Stage 4 fleet. Nine of these involved a light vehicle tag on the main ramp below the boundary. The system logged these events but nobody saw them, because the standing report covered only alert counts and system health. Both the operator and the supervisor raised the suppression without being asked. The argument is that a silent system had taught the operator to treat silence as meaning the road was clear, and that he carried this expectation over the crest. That mechanism is inferred from his account and the alert pattern rather than measured. On that basis it is likely, not more than likely, that the suppression influenced the occurrence.

The finding is not that a geofence existed. It is that the boundary moved 180 m and nobody outside the group that moved it was told. The suppression is therefore treated as a causal factor, with the qualification that its influence works through the operator's expectation rather than through the alert that was not given.

Positive radio calls required only at listed intersections, and the list is rewritten once a year

Clause 4.3.1 of BIN-TMP-004 requires a positive call "before entering any intersection listed at Appendix B". Clause 4.3.2 provides for intersections not listed that "No call is required."

An intersection reaches Appendix B only if it appears on the Schedule 1 road layout drawing. That drawing is reissued only at the annual review. No provision allows an intersection to be added between reviews, and none requires a new road to be referred to the plan's owner. The RL 312 spur appeared on neither document. It is virtually certain that the control was framed in this way, because the plan text, its appendix and its review history are all documentary.

The practice on site matched the wording. The truck operator, the light vehicle driver and the night shift supervisor each independently described calls being made at the nine intersections on the crib room board and nowhere else. The pit channel recording for the period contains no call about the spur, the ramp or a light vehicle in the pit.

Had a call been required, the conditions for it to work were in place. The channel was open, both vehicles were on it, and at 02:46:18 the dispatcher placed HT-407 as "about four minutes off" the loading face. The one remaining uncertainty is that a call made and not acknowledged still leaves a driver deciding whether to proceed. It is very likely that the absence of a requirement influenced the occurrence.

This was not an isolated gap. Eleven of the fourteen new Stage 4 roads cut since the last review were never referred, and no out-of-cycle review has been held in the plan's eleven years. The earliest date on which the spur could have been listed was 12/11/2026, three months after the collision. The way the spur came to be opened is argued under Organisational Influences, and the guidance that left the review trigger undefined is argued under External Influences.

The restriction of positive calls to listed intersections, with no route for adding a new intersection between annual reviews, was a causal factor (Safety issue).

One emergency response paramedic rostered on night shift, with the second responder called from the village

The night shift was rostered with one qualified mines rescue paramedic at the surface medical facility. The second responder, who must be present before any extrication begins, is called in from the accommodation village 11 km from the gatehouse. On the night, the sequence was as follows:

- dispatch paged the team at 02:48
- the first response vehicle reached the scene at 02:54
- the second responder arrived with the hydraulic rescue tool at 03:07
- the driver was extricated at 03:14, 27 minutes after the collision.

The post-incident review found that extrication was delayed by approximately 9 minutes. It is virtually certain that this arrangement existed, since the roster, the response log and the review agree on the times.

The collision and the injuries were complete before the team was paged. The review recorded that the delay did not affect the clinical outcome, because the injuries were stable and the driver was

conscious throughout. It is unlikely that the arrangement influenced this occurrence or its outcome, and it is not a causal factor.

It remains important. The driver was restrained and protected by an intact roll-over structure. With a fuel fire, an unconscious casualty or a compromised airway, the same nine minutes could produce a different outcome. The site's night shift extrication capability is one person plus a drive from the village, and nothing in the arrangement changes with the severity of the casualty. It is very likely that this increased risk, and it is reported as a safety issue that increased risk.

Local Conditions

The spur meets the ramp beyond a crest, giving 41 m of sight line against a plan minimum of 120 m

The mine survey department picked up the intersection to the eye heights of both vehicles. The spur meets the ramp 34 m beyond the high point of a crest. An operator descending at an eye height of 5.80 m first had line of sight to the spur mouth at 41 m. A driver at the spur mouth, at an eye height of 1.45 m, first had line of sight to a descending truck at 38 m, and the 1.6 m inner windrow hid the truck's lower body for the last 25 m of the spur's approach. Seven days earlier, the dewatering technician had independently described the same condition: "you cannot see up the ramp from that road." It is virtually certain that the sight line was 41 m.

The significance of that figure lies in its relationship to stopping distance. BIN-TMP-004 clause 6.2 requires the available sight distance at every intersection to be not less than the stopping distance of the largest vehicle at the posted speed, and in no case less than 120 m. Where that cannot be achieved, the clause requires a stop, signalised or controlled intersection. The stopping distance of a 220 t class truck travelling empty at the posted 40 km/h on a 9 per cent descending grade is 96 m. This figure was calculated with the manufacturer's data and a 1.5 second response. The intersection therefore met neither limb of clause 6.2. The available sight line was less than half the distance the truck needed to stop.

At the speed recorded on the download, the geometry left about four seconds between first sighting and the intersection. Because clause 6.4 applies only to intersections shown on the plan's road layout drawing, and the RL 312 spur was not on it, the intersection carried no warning sign, no give-way control, no delineation and no posted speed. Nothing on the approach compensated for the deficit.

The investigation considered whether the sight line could have played no part. A restricted sight line only matters once a vehicle is in it, and the presence of the light vehicle at the spur mouth depended on other factors. Those factors are argued under Organisational Influences and Individual Actions. They do not alter what the geometry imposed once both vehicles were converging. At 41 m, a truck at the posted speed could not stop for anything at this intersection, however attentive its operator. The operator's attention, the light vehicle driver's training and any radio call all operated within a margin the geometry had already removed. The condition followed from the spur being cut and opened under a works order without a management of change, which is argued under Organisational Influences.

The 41 m sight line very likely led to both vehicles arriving at the intersection with neither operator aware of the other, and it was a causal factor in the collision.

Dust on the ramp reduced the truck operator's ability to see the light vehicle

This matter was examined and could not be settled.

Some evidence pointed towards dust:

- The water cart's last pass on that section was at 01:10, 1 hour 37 minutes before the collision. This was against a standing 90 minute interval, so a pass was 7 minutes overdue.
- The fixed dust monitor at the ramp crest, 180 m up-ramp of the intersection, logged PM10 at 142 micrograms per cubic metre in the 02:40 to 02:50 interval, against a shift mean of 88.
- The truck operator described dust hanging on the ramp after the cart's earlier pass.
- The truck was descending behind its own lights into a pit with no fixed lighting.

Other evidence pointed the other way:

- The light vehicle driver had been in the same place about four seconds earlier, and he was firm that the air was clear and that he could see the ramp surface and the windrow.
- The truck operator declined to say that the dust made any difference.
- The reading of 142 was well below the site trigger of 250. It measured particulate at a monitor head 180 m away, not what could be seen from either cab.

The strongest point against the hypothesis is the sighting distance itself. The truck operator saw the light vehicle at 41 m, which is the distance the survey gives for the geometry in clear air. That leaves dust nothing to explain.

The deviation in watering and particulate was real, and it occurred at the right place and time. Whether it affected detection was about as likely as not. Resolving the question would require a measurement of visibility from either cab on the night. No such measurement exists, and nothing at Bindarra measures visibility. The matter remains unresolved, and nothing from it is carried into the findings.

The ramp surface was wet and the truck could not stop in the distance available

It was very unlikely that the ramp was wet. No precipitation was recorded, the last rainfall was 26 days earlier, and the investigator found the ramp dry throughout at 06:20. In addition, the 62 m braked scuff showed no locked or sliding wheel, and the mean deceleration of 1.85 m/s² matched the unit's tested dry figure of 1.9 m/s². The hypothesis was therefore excluded.

Night 10 of 14, in the eleventh hour of a 12.5 hour night shift, in the circadian low

It is virtually certain that the truck operator was on night 10 of a 14 night roster. At 02:47 he was 10 hours 47 minutes into a shift worked from 18:00 to 06:30, within the period of lowest circadian alertness. Every limit in the contractor's fatigue management standard had been met. His own sleep record showed 6 hours 30 minutes of sleep in the preceding day.

Fatigue would have had to act through a slow or absent response, and the recorded data contradicts that. The interval from the last steady data point to the rise in brake pressure was 1.4 seconds. This is inside the 1.5 second response assumed in the stopping distance calculation. The braking and steering that followed were immediate and correct. Even a response time of zero, at 41 m and 37 km/h, would still have ended in a collision. It is therefore unlikely that fatigue influenced this occurrence, and it is not a causal factor.

It remains likely to be important as a standing exposure. The same roster is worked by every haul truck operator in the Stage 4 pit and on three other pits on the site, in an unlit pit where the operator's own eyes are the last defence. It is carried forward as a contributory factor of that standing, risk-increasing kind, not as a factor in this collision.

Individual Actions

Neither operator made a positive call on the pit channel before entering the intersection

The pit channel was logged continuously, and the investigator listened to the whole period from the return from crib at 02:22:41 to two minutes after the collision. It contains no transmission by either vehicle about the RL 312 spur, the dewatering bore, or a light vehicle being anywhere in the Stage 4 pit. LV-118 made no transmission that shift after logging on at 18:04. The truck operator, the light vehicle driver and the night shift supervisor were interviewed separately and without being led, and each described the same practice: calls are made at the nine intersections on the crib room board and at no others. The supervisor stated that he had never corrected anybody for not calling elsewhere. He also considered that requiring calls everywhere would make the channel unusable. It is virtually certain that no call was made.

That practice matched the plan exactly. Clause 4.3.2 of BIN-TMP-004 provides that at intersections not listed at Appendix B, vehicles "shall give way in accordance with clause 4.1. No call is required." The RL 312 intersection was not listed. The omission therefore departed from no rule, instruction or local custom known to either operator, and it is not attributed to either of them. The question of why the intersection was absent from Appendix B is argued under Risk Controls.

A call from either vehicle would have been heard by the other, by dispatch and by the supervisor. The dispatcher was already tracking the truck closely enough to tell the excavator operator at 02:46:18 that "407's about four minutes off you". A call would therefore have put each vehicle in the other's knowledge. It would not by itself have given them separation, because the two still had to resolve who went first across 41 m of sight line. For that reason the influence is put at likely rather than higher.

The absence of a positive call on the pit channel likely contributed to both vehicles arriving at the intersection with neither operator aware of the other, and it is a causal factor.

Light vehicle entered the ramp from the spur without stopping and without establishing the truck's position

The driver's first account was that he had stopped at the mouth. On read-back he corrected this himself, against his own interest. His corrected account is that he came out of the spur at about 25 km/h, looked left up the ramp, saw nothing and drove out, and that he does not believe he came to a complete stop. The scene evidence agrees: the impact was 6.2 m into the ramp from the eastern edge of the spur mouth, and the gouge runs from that point. It is virtually certain that he entered the ramp in this way.

He described himself as applying clause 4.1, which requires light vehicles to give way to heavy vehicles at all times. Nothing at the mouth told him more. There was no give-way control, stop line, warning sign or posted speed, because clause 6.4 applies only to intersections on the road layout drawing. He had done no Bindarra pit driving assessment and had never seen the traffic management plan. The reasons for that are argued under Organisational Influences.

The survey bears directly on whether stopping would have changed what he could see. From the mouth, at an eye height of 1.45 m, a driver "first has line of sight to a descending truck at 38 m". The inner windrow also hides the truck's lower body for the last 25 m of the approach. A look left from a stationary position would therefore have shown him approximately what his look at 25 km/h showed him. The truck needed 96 m to stop, and a full stop would have gained a few metres of sight line at most.

Without the light vehicle on the ramp there would have been no collision, and entering without establishing the truck's position was the proximate step. Because the geometry did not allow the truck's position to be established by looking, the influence is put at very likely rather than certain.

Entering the ramp without stopping and without establishing the truck's position very likely contributed to both vehicles arriving at the intersection unaware of each other. It is a causal factor, taken in an intersection where a stop would not have provided the sight distance needed.

Truck operator braked and steered into the windrow within 1.4 seconds of sighting

The fleet management system download records the last steady data point at 02:47:06, at 37 km/h and 41 m above the intersection. Service brake pressure rose from 0 to 620 kPa in 0.4 seconds from 02:47:07.4. In the same movement the operator took the retarder to maximum and applied 14 degrees of right steering. That is a response of 1.4 seconds, inside the 1.5 seconds assumed in the plan's own stopping distance calculation. Mean deceleration to the collision was recorded as 1.85 m/s², against 1.9 m/s² for this unit at its brake test on 03/08/2026. The scene marks corroborate the data, with the braked scuff deviating 1.4 m to the right over its first 20 m.

Two consequences follow. The impact speed was 24 km/h, where with no braking it would have been 37 km/h. The steering took the collision towards the wall side of the road, the only direction that did not carry the light vehicle towards a 100 m drop, and LV-118 came to rest against the inner windrow. After stopping 19 m below the intersection, the operator reported the collision at 02:47:41. He remained in the cab as instructed until 02:48:44, then went down and stayed with the driver until the response team arrived.

The operator's prompt braking and steering very likely reduced the severity of the outcome, and it is recorded as a positive action rather than as a causal or contributory factor.

Occurrence Events

Both vehicles arrived at the intersection with neither operator aware of the other

Every line of enquiry in this analysis arrives at the same condition, so it is treated here as a single mechanism rather than as a series of separate failures.

The haul truck. HT-407 was descending the ramp empty at 37 km/h. Its in-cab road network showed no intersection at RL 312, and there was no warning sign on the approach. No call had been made on the channel. The collision awareness system logged the light vehicle at 21.4 m but presented nothing in the cab.

The light vehicle. LV-118 was leaving the spur at about 25 km/h. There was no give-way control at the mouth, no call had been made, and its driver had no knowledge of the truck.

The sight lines. The survey established that a descending operator first has line of sight to the spur mouth at 41 m. A driver at the spur mouth first has line of sight to a descending truck at 38 m. The download placed the truck's last steady data point 41 m above the intersection at 02:47:06. This is the sighting distance the geometry predicts, and it left about four seconds.

Awareness before the sighting. None of the following contained any indication that either operator knew of the other before that moment:

- the pit channel recording
- the download
- the collision awareness log
- the three interviews.

The night shift supervisor did not know a light vehicle was in the pit. There was no permit, call-in or board by which he could have known. The light vehicle's movements before the last two seconds rest on the driver's own account alone, which is why this condition is put at extremely likely and not higher.

The reasons each source of awareness was absent are argued under Risk Controls, Local Conditions and Individual Actions. The possible part played by dust on the ramp could not be resolved and is set out under Local Conditions. Once the two vehicles were converging unseen at 41 m, nothing between the convergence and the collision was capable of preventing it.

It was extremely likely that both vehicles arrived at the intersection with neither operator aware of the other, and extremely likely that this led to the collision, which makes it a causal factor.

The haul truck's front left struck the right side of the light vehicle at 24 km/h

The download recorded the brake application at 02:47:07.4 and a deceleration spike at 02:47:12.0, 4.6 seconds later, at 24 km/h. The truck's accelerometer logged 1.9 g longitudinal at the moment of impact.

The scene examination placed the impact 6.2 m into the ramp from the eastern edge of the spur mouth and 3.1 m out from the inner windrow toe. Damage to LV-118 was concentrated on its right side from the B pillar forward. Damage to HT-407 was confined to the front left bumper, the left mudguard and the lower two steps of the access ladder, with none above the bumper line. Right-side glass and mirror fragments lay from 4.8 m to 14 m down-ramp of the impact position. The recorded data, the damage pattern and the debris field all agree.

The light vehicle was struck on its side, not its front. At an intersection where a vehicle emerging from the spur meets a truck already committed to the ramp, the geometry made a side impact the inevitable form of the collision. A side impact from a truck of the 220 t class left the outcome to three things: the light vehicle's structure, its restraint and the impact speed. The speed had been reduced from 37 km/h to 24 km/h by the truck operator's braking, which is discussed under Individual Actions.

It was virtually certain that the truck's front left struck the right side of the light vehicle at 24 km/h, and virtually certain that this impact was the means by which the convergence became an injury, which makes it a causal factor.

The light vehicle was pushed 18 m down the ramp and rolled onto its left side

A continuous gouge ran 18.4 m down-ramp from the impact position. At 11.2 m it changed from a tyre scuff to a body panel gouge, which marks where the vehicle went over. LV-118 came to rest on its left side against the inner windrow, facing down-ramp. The scene measurements, the vehicle examination and the medical record agree.

The driver sustained a fractured left clavicle, three fractured left ribs and a small pneumothorax, and was conscious throughout. The door on the resting side was against the ground, so extrication was through the rear of the vehicle.

Two defences worked:

- The roll-over protection structure was deformed inward at the right rear mounting but was not breached.
- The driver's seatbelt was found fastened, with loading marks on the webbing.

These are the injuries a side impact and a roll-over produce when the structure holds and the restraint is worn. Without either of them, the same sequence would have had a materially different outcome. The emergency response that followed is considered under Risk Controls.

It was virtually certain that the light vehicle was pushed down the ramp and rolled onto its left side, and virtually certain that the roll-over was how the collision produced the injuries, which makes it a causal factor.

The haul truck's service brakes were degraded and could not achieve the required stopping performance

The hypothesis that the truck's service brakes were degraded was ruled out as very unlikely, for these reasons:

- The brake performance tests on 03/08/2026 and 14/08/2026 gave mean decelerations of 1.9 m/s² and 1.92 m/s².
- The occurrence download gave 1.85 m/s², between the two tests and within normal scatter.
- The pre-start check sheet for 11/08/2026 recorded no defects.
- No brake, retarder or park brake defect had been raised against the unit in the preceding twelve months.

Conclusions

The investigation identified the following causal and contributory factors. They should not be read as apportioning blame or liability to any organisation or individual.

A finding stated without a likelihood term was established as fact.

Findings

1. Mines Safety Guideline MSG-14 required the traffic management plan to be reviewed after any significant change to the road network without defining the term. It left the operator to nominate its own interaction zones with no minimum and no basis for the nomination, and it allowed light vehicle detectability by any one of five alternative means. Bindarra's plan met the guideline on its face, and the mine was not in breach of it. (Safety issue)

2. The 240 m RL 312 dewatering spur was cut under works order WO-26-4471, which classified the work as routine earthworks within the approved Stage 4 design and recorded management of change as not required under CRL-PRO-022 clause 5.3. As a result, none of the three referrals at clause 5.4 was made, to the traffic management plan owner, to the survey department for the in-cab road network, and to the contractor's technical services group for the collision awareness system geofences. Eleven of the fourteen roads cut in Stage 4 since the last plan review were built under works orders in the same way, and none of them was referred to the plan's owner. (Safety issue)

3. It is very likely that the light vehicle driver's Bindarra site-specific supplement and Bindarra light vehicle pit driving assessment were never verified before his access card was issued. CRL-STD-009 clause 7.4 states that neither may be substituted, but the gatehouse issued cards against the contractor's recognition agreement alone, and the access card system held no field for either item. (Safety issue)

4. On 09/06/2026 the S4-LOAD suppression geofence was extended northward to a boundary 180 m up-ramp of the RL 312 intersection, which placed the intersection inside the suppressed area. The extension was not recorded in the traffic management plan or reviewed against it. At 02:47:09.2 the collision awareness system on HT-407 logged LV-118 at 21.4 m, closing at 13 km/h, and suppressed the alert. In the preceding twelve months, 68 proximity events inside the geofence were suppressed, nine of them involving a light vehicle tag on the main ramp. None was reported, because the standing report did not include suppressions. (Safety issue)

5. BIN-TMP-004 required a positive call on the pit channel only at the nine intersections listed at Appendix B. An intersection could be added to that list only through a road layout drawing that was reissued at the annual review. No provision required a new road to be referred to the plan's owner when it was cut. The RL 312 spur was not listed, and the next date on which it could have been listed was 12/11/2026. (Safety issue)

6. The Bindarra night shift was rostered with one mines rescue paramedic. The second responder, who had to be present before any extrication, was called in from the accommodation village 11 km from the gatehouse. The post-incident review found that the extrication was delayed by approximately 9 minutes and that the delay did not affect the clinical outcome on this occasion. (Safety issue)

7. The spur met the ramp 34 m beyond the high point of a crest. This gave the truck operator 41 m of sight line, against a stopping distance of 96 m at the posted speed and a plan minimum of 120 m. The intersection carried no warning sign, no give-way control, no delineation and no posted speed, because clause 6.4 applied only to intersections on the road layout drawing. The same sight

line had been reported in hazard report HAZ-2026-0917 seven days before the collision, and that report was closed with a reminder to the crew.

8. The investigation could not determine whether dust on the ramp reduced the truck operator's ability to see the light vehicle, and the evidence was evenly balanced. Several matters pointed towards an effect:

- The water cart pass was 7 minutes overdue.
- The ramp crest monitor read 142 micrograms per cubic metre of PM10, against a shift mean of 88.
- The truck operator described dust hanging on the ramp.

Several matters pointed against an effect:

- The light vehicle driver was firm that the air was clear.
- The truck operator declined to say that the dust made a difference.
- The reading was below the site trigger of 250 and was taken 180 m away.
- The sighting at 41 m matched what the geometry gives in clear air.

The matter could not be resolved because nothing at Bindarra measured visibility from either cab.

9. The ramp surface was dry. No rain had fallen for 26 days, the investigator found no damp patches at or above the intersection, the braked scuff showed no sliding wheel, and the deceleration achieved was 1.85 m/s² against the unit's tested 1.9 m/s² on a dry grade.

10. The truck operator was on night 10 of a 14 night swing, 10 hours 47 minutes into a 12 hour 30 minute shift and within the window of lowest circadian alertness. Every limit in the contractor's fatigue management standard had been met. His recorded response of 1.4 seconds was inside the 1.5 seconds assumed in the plan's stopping distance calculation.

11. Neither vehicle made any transmission on the pit channel about the RL 312 spur, the dewatering bore or a light vehicle being in the Stage 4 pit. No call was required at an unlisted intersection under clause 4.3.2, and the truck operator, the light vehicle driver and the night shift supervisor each independently described a practice that matched that clause.

12. The light vehicle left the spur at about 25 km/h without coming to a complete stop, and it entered the ramp without its driver having established the truck's position. From the spur mouth the survey gave 38 m of sight line, and the windrow screened the truck's lower body, so a look from a stopped position would have shown approximately what a look at 25 km/h showed.

13. The truck operator applied the brakes, took the retarder to maximum and applied 14 degrees of right steering within 1.4 seconds of the last steady data point. This reduced the impact speed from 37 km/h to 24 km/h and kept the collision against the inner windrow.

14. It is very likely that both vehicles arrived at the intersection with neither operator aware of the other until the first sighting available, at 41 m and about four seconds. The night shift supervisor did not know that a light vehicle was in the pit.

15. At 02:47:12 the front left of HT-407 struck the right side of LV-118 at 24 km/h, 6.2 m into the ramp from the eastern edge of the spur mouth.

16. LV-118 was pushed 18.4 m down the ramp and rolled onto its left side against the inner windrow. Its roll-over protection structure was deformed but not breached, and the driver's seatbelt was fastened and showed loading marks. The driver sustained a fractured left clavicle, three fractured left ribs and a small pneumothorax.

17. The service brakes of HT-407 performed to specification. Tests on 03/08/2026 and 14/08/2026 gave 1.9 m/s² and 1.92 m/s², the occurrence download gave 1.85 m/s², and no brake defect had been raised against the unit in the preceding twelve months.

Causal factors

- Collision awareness alerts were suppressed inside a geofence that had been extended to cover the new intersection, and this is virtually certain. It is likely that the suppression contributed to the collision. The mechanism was not the advisory alert itself, which would have sounded 1.8 seconds after braking began. It was the absence of any alert on that ramp over the truck operator's time at the mine, which had led him to read silence as an empty road. The evidence also supports a reading in which the suppression was a serious deficiency in the fleet's defences rather than a cause of this collision. (Safety issue)
- It is virtually certain that MSG-14 left the review trigger undefined and the nomination of interaction zones to the site. It is likely, though on a firmer balance than for none of the other causal factors, that this contributed to the occurrence, because it gave the roads that were not referred to the plan's owner no defined point at which they had to be considered. The plan's own clause 9.1 allowed a review at any time. The finding concerns the breadth of the discretion and not any failure by the State Mines Inspectorate. (Safety issue)
- That the access spur was cut and opened to traffic under a works order, with no management of change raised, is virtually certain. It is very likely that this contributed to the occurrence, because a management of change record was the only route by which the new intersection could have reached the traffic management plan, the in-cab road network or the geofence configuration. (Safety issue)
- With virtual certainty, the spur met the ramp beyond a crest, giving 41 m of sight line against a plan minimum of 120 m and a stopping distance of 96 m. It is very likely that this geometry contributed to the collision by leaving about four seconds between first sighting and the intersection.
- The requirement for positive radio calls only at listed intersections, with the list rewritten once a year, is virtually certain. It is very likely that this requirement contributed to the occurrence, because the pit channel was open, both vehicles were on it, and a call at the spur was not required by the plan's own terms. (Safety issue)
- Neither operator made a positive call on the pit channel before entering the intersection, and this is virtually certain. It is likely that the absence of a call contributed. That finding is tempered by two points. A call would have given each operator knowledge of the other but not separation over 41 m of sight line. The omission also matched both the plan and the practice described by everyone on the shift.
- It is very likely that the contractor induction was accepted as equivalent and that the site supplement and pit driving assessment were never verified. It is likely that this contributed to the occurrence. The finding is tempered by two points. The give-way rule the assessment would have taught was one the driver was already applying. The intersection's sight line also defeated attention on its own. (Safety issue)
- The light vehicle entered the ramp from the spur without stopping and without establishing the truck's position, which is virtually certain. It is very likely that this contributed to the collision, although the survey showed that a full stop would have given the driver only a few more metres of sight line and not the 96 m the truck needed.

- Both vehicles arrived at the intersection with neither operator aware of the other, which is very likely. It is very likely that this convergence contributed to the collision, which followed from it directly.
- The haul truck's front left struck the right side of the light vehicle at 24 km/h, which is virtually certain. It is virtually certain that this impact contributed to the occurrence.
- Following the impact, the light vehicle was pushed 18.4 m down the ramp and rolled onto its left side, which is virtually certain. It is virtually certain that this contributed to the injuries the driver sustained.

Contributory factors

Neither factor below was found to have influenced the occurrence or its outcome. Each is reported because it increased risk.

- It is virtually certain that the night shift was rostered with one emergency response paramedic and that the second responder was called from the village. It is very likely that this arrangement increased risk, because the night shift's extrication capability was a single person plus a drive of 11 km and did not change with the severity of the casualty. (Safety issue)
- The truck operator was on night 10 of 14, in the eleventh hour of a 12.5 hour night shift, in the circadian low, and this is virtually certain. It is likely that this roster increased risk as a standing exposure that the Stage 4 night roster places on every haul truck operator, in an unlit pit where an operator's own eyes are the sole defence at an unlisted intersection.

Other findings

- The hypothesis that the ramp surface was wet and that the truck could not stop in the distance available was examined and ruled out. The weather data, the surface as examined and the deceleration achieved all showed a dry road.
- The hypothesis that the haul truck's service brakes were degraded and could not achieve the required stopping performance was examined and ruled out. Brake tests before and after the occurrence, and the occurrence download itself, agreed within normal scatter.
- The truck operator braked and steered into the windrow within 1.4 seconds of sighting, and this was found to reduce risk. The action lowered the impact speed from 37 km/h to 24 km/h and kept the light vehicle against the inner windrow rather than towards the drop on the outer side of the ramp.

Safety action and recommendations

Traffic management guidance leaves the review trigger undefined and the interaction zones to the site

Mines Safety Guideline MSG-14 revision 4 is made under the Mining Operations Safety Regulations 2018 r 6.22. It requires a mine to do three things:

- review its traffic management plan after any significant change to the road network,
- nominate the intersections and interaction zones at which positive communication is required, and
- make a light vehicle detectable by a heavy vehicle operator by any one of five means.

It is virtually certain that the guideline defines none of these three terms and sets no minimum or basis for nominating interaction zones. It also provides that the plan is a site document, not approved by or submitted to the Inspectorate except on request after an occurrence. Bindarra's plan, BIN-TMP-004, met the guideline on its face, and the mine was not in breach. The safety issue lies in the width of the discretion the guideline leaves, not in any failure by the Inspectorate.

It is likely that this discretion contributed to the occurrence. Review of the plan and nomination of interaction zones rested entirely on a single site document, reviewed once a year, with no outside eye on it. Eleven of the fourteen new roads cut in Stage 4 since the last review were never referred to the plan's owner. This finding was reached on the balance and is held with less confidence than the other causal findings. The guideline did not prevent the mine from holding an out-of-cycle review under its own plan, or from listing every intersection it had.

No safety action has been recorded in response to this safety issue, and no safety recommendation is made.

Access spur cut and opened to traffic under a works order, with no management of change raised

The access spur to bore DW-14 at RL 312 was cut and opened to traffic under works order WO-26-4471 without a management of change record. CRL-PRO-022 clause 5.2 requires such a record before any change to the mine's road network. Clause 5.3 excludes routine earthworks under a works order within an approved pit design, and the works order relied on that exclusion. As a result, none of the three referrals at clause 5.4 was made: to the traffic management plan owner, to the survey department for the fleet management system road network, and to the contractor's technical services group for the collision awareness system geofences. This was a causal factor in the occurrence.

The issue extends beyond the RL 312 spur. Fourteen new roads have been cut in Stage 4 since the last plan review. The three raised under a management of change record were referred to the plan's owner. None of the eleven raised under a works order was.

Safety action taken

No safety action addressing this issue has been recorded. After the occurrence, a site-wide direction required a positive call at every Stage 4 intersection, listed or not. That direction answers the radio call requirement. It does not change how new roads are classified or referred.

Safety recommendation

It is recommended that [NOT ESTABLISHED: the organisation responsible for CRL-PRO-022, by role] ensure that earthworks creating a new intersection with a haul road generate the three

referrals held at clause 5.4. This may be achieved either by removing such earthworks from the exclusion at clause 5.3, or by carrying the three referrals on the works order form at Schedule 4.

Contractor induction accepted as equivalent, with the site supplement and pit driving assessment never verified

CRL-STD-009 permitted a contractor's general induction to be accepted as equivalent under a recognition agreement. It excluded from that equivalence the Bindarra site-specific supplement and the Bindarra light vehicle pit driving assessment, and it required the engaging department to verify both before a site access card was issued. It is extremely likely that this verification did not take place for the light vehicle driver.

The gatehouse issued cards against the recognition agreement alone. It held no record of either item and was not asked for one. The access card system did not distinguish a person who could drive in an operating pit from one who could not. The standard set out the correct requirement. The weakness was that the card issue process had no point at which the verification could be made.

The issue matters because the pit driving assessment is where a light vehicle driver is taught how the pit's traffic rules work in practice, including give-way, positive calls and conduct at a blind intersection. On his third night at the mine, the driver was on an active haul road without an assessment that the site's own standard states may not be substituted. It is likely that this contributed to the light vehicle entering the ramp from the spur without stopping and without establishing the truck's position. That influence was tempered for two reasons. The give-way rule he would have been taught was one he was already applying, and the sight line at the intersection was by itself sufficient to defeat attention.

No safety action has been recorded in response to this safety issue, and no recommendation is made.

Positive radio calls required only at listed intersections, and the list is rewritten once a year

The traffic management plan, BIN-TMP-004, required a positive call on the pit channel only before entering an intersection listed at Appendix B. An intersection could reach Appendix B only through the road layout drawing at Schedule 1. That drawing was reissued once a year at the annual review, and nothing in the plan required a new road to be referred to the plan's owner when it was cut. The RL 312 spur appeared on neither document, and eleven of the fourteen new Stage 4 roads cut since the last review had not been referred. The mine's own defence against this interaction was therefore unavailable at this intersection by the plan's own terms. The earliest date on which the spur could have been listed was 12/11/2026, three months after the collision.

Safety action taken

At 05:10 on 12/08/2026, two hours twenty-three minutes after the collision, the Mining Manager of Corella Resources issued a site-wide direction. It requires a positive call on the pit channel before entering any intersection in the Stage 4 pit, whether or not it is listed in the plan, until further notice. The direction was read at the following four pre-starts and posted beside the Appendix B board in the Stage 4 crib room. It remains in force until BIN-TMP-004 is amended. This action addresses the issue in part. It restores the call at every Stage 4 intersection, but it does not change the route by which an intersection reaches Appendix B, and it does not require new roads to be referred.

Safety recommendations

It is recommended that the operator amend BIN-TMP-004 to achieve two outcomes. An intersection must be able to reach Appendix B between annual reviews. A new road must be referred to the plan's owner when it is opened to traffic.

Collision awareness alerts suppressed inside a geofence extended to cover the new intersection

The collision awareness system on the Stage 4 haul fleet gave operator advisory alerts only. It suppressed them inside the S4-LOAD geofence, which was drawn around the loading face to stop nuisance alerts. On 09/06/2026 the contractor's technical services group extended the boundary northward to a point 180 m up-ramp of the RL 312 intersection. This was done as a system configuration task, without recording it in the traffic management plan or reviewing it against that plan, and nobody outside the group was told. At 02:47:09.2 the system detected the light vehicle at 21.4 m and raised no alert.

The safety significance lies in what operators had come to expect, more than in the alert that did not sound. The suppressed alert would have come after braking had begun, but months of silence on the ramp had likely taught the truck operator that silence meant an empty road. No training record on site, for any person, covered the geofences.

Safety action taken

On 28/08/2026 Warrigal technical services added suppressed proximity events to the monthly collision awareness report, broken down by geofence and by vehicle class. It also re-cut the previous twelve months on the same basis for the Stage 4 fleet. Before this, the 68 suppressed events inside S4-LOAD, nine of them a light vehicle on the main ramp, had been recorded but never reported. The first report is due on 30/09/2026. This action makes suppressions visible but does not control changes to the boundary or inform operators, so it addresses the issue in part.

Safety recommendation

It is recommended that [NOT ESTABLISHED: organisation to which the recommendation is addressed] ensures that any change to a suppression geofence boundary is treated as a change to a documented control under CRL-PRO-022 clause 5.2. It is also recommended that operators are told in writing where the collision awareness system is suppressed and why.

One emergency response paramedic rostered on night shift, with the second responder called from the village

The Bindarra night shift was rostered with one qualified mines rescue paramedic, based at the surface medical facility. The second responder, who had to be present before any extrication could begin, was called in from the accommodation village 11 km from the gatehouse. On the night of the occurrence, dispatch paged the team at 02:48 and the first response vehicle reached the scene at 02:54. The second responder arrived with the hydraulic rescue tool at 03:07, and the driver was extricated at 03:14, 27 minutes after the collision. The site's post-incident review found that the extrication was delayed by approximately 9 minutes while the second responder and the tool were awaited.

The collision and the driver's injuries were complete before the emergency response was paged. It was therefore unlikely that the response time affected the occurrence or its outcome. The post-incident review reached the same conclusion on clinical grounds, because the driver was conscious and his injuries were stable throughout.

The arrangement nonetheless very likely increased risk. It left the site's night shift extrication capability resting on a single person and a drive from the village, and nothing in the arrangement

changed with the severity of the casualty it was called to. A fuel fire, an unconscious casualty, or an airway requiring the vehicle to be opened within the first ten minutes would each turn the same delay into a different outcome.

No safety action has been recorded in response to this safety issue, and this report makes no safety recommendation on it.

Distribution

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Mining Manager and owner of BIN-TMP-004	H. Quaile
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